

Sample AI Knowledge Document

This document is intended for testing question answering over uploaded PDFs. It contains information on artificial intelligence, machine learning, and data science.

Section 1: Artificial Intelligence

Artificial Intelligence (AI) is the field of creating systems that perform tasks normally requiring human intelligence. Examples include language understanding, image recognition, planning, and decision making.

Section 2: Machine Learning

Machine Learning (ML) is a subset of AI in which algorithms learn patterns from data. Common learning paradigms include supervised, unsupervised, and reinforcement learning.

Section 3: Deep Learning

Deep Learning uses neural networks with many layers. Popular architectures include Convolutional Neural Networks (CNNs) for images and Transformers for language tasks.

Section 4: Data Science

A typical data science workflow includes data collection, cleaning, exploratory data analysis, feature engineering, model training, evaluation, and deployment.

Section 5: Key Facts

Python is one of the most widely used programming languages for AI. Popular libraries include NumPy, pandas, scikit-learn, TensorFlow, and PyTorch. FastAPI is commonly used to build AI APIs.